and the conditions that the curve may be an ellipse, are

$$a+b=+,\qquad ab-h^2=+,$$

the former of which requires no demonstration; to prove the latter, changing merely the variables under the integral sign, I write

$$a'=\int x'^2\,dm',\quad b'=\int y'^2\,dm',\quad h'=\int x'y'\,dm',$$

these quantities being of course respectively equal to a, b, h, we have then

$$ab'+a'b-2hh'=\iint (xy'-x'y)^2\,dm\,dm'=2(ab-h^2),$$

or since the quantity under the integral sign is a square, $ab-h^2$ is positive.

For the analogous problem *in solido*, we have

$$a=\int x^2\,dm,\ b=\int y^2\,dm,\ c=\int z^2\,dm,\ f=\int yz\,dm,\ g=\int zx\,dm,\ h=\int xy\,dm,$$

and it is to be shown that the equations

$$(a,b,c,f,g,h\mid p,q,r)^2=1,$$
$$(a+b+c)(p^2+q^2+r^2)-(a,b,c,f,g,h\mid p,q,r)^2=1,$$

represent respectively ellipsoids.

The conditions in the first problem are

$$\begin{aligned} a+b+c&=+,\\ bc+ca+ab-f^2-g^2-h^2&=+,\\ abc-af^2-bg^2-ch^2+2fgh&=+,\end{aligned}$$

the first of which is obviously true: as regards the second, the theorem *in plano* shows that each of the quantities $bc-f^2$, $ca-g^2$, $ab-h^2$ is positive, or merely reproducing the investigation, we find

$$2(bc+ca+ab-f^2-g^2-h^2)=\iint\left[(yz'-y'z)^2+(zx'-z'x)^2+(xy'-x'y)^2\right]dm\,dm',$$

which proves the theorem, and where it is to be observed that the integral may also be written

$$\iint\left[(x^2+y^2+z^2)(x'^2+y'^2+z'^2)-(xx'+yy'+zz')^2\right]dm\,dm';$$

and for the third, we find in a precisely similar manner,

$$6(abc-af^2-bg^2-ch^2+2fgh)=\iint\begin{vmatrix} x & y & z\\ x' & y' & z'\\ x'' & y'' & z''\end{vmatrix}^2 dm\,dm',$$

which proves the theorem. The integral may also be written

$$\iint \begin{vmatrix} x^2+y^2+z^2 & xx'+yy'+zz' & xx''+yy''+zz'' \\ x'x+y'y+z'z & x'^2+y'^2+z'^2 & x'x''+y'y''+z'z'' \\ x''x+y''y+z''z & x''x'+y''y'+z''z' & x''^2+y''^2+z''^2 \end{vmatrix} dm\,dm'.$$

The conditions in the second problem are

$$\begin{aligned} (b+c)+(c+a)+(a+b) &= +, \\ (c+a)(a+b)+(a+b)(b+c)+(b+c)(c+a)-f^2-g^2-h^2 &= +, \\ (a+b)(b+c)(c+a)-(b+c)f^2-(c+a)g^2-(a+b)h^2-2fgh &= +, \end{aligned}$$

the first and second of which are respectively equivalent to

$$a+b+c=+,$$

$$(a+b+c)^2+bc+ca+ab-f^2-g^2-h^2=+,$$

which are already proved. The last may be written

$$(a+b+c)(bc+ca+ab-f^2-g^2-h^2)-(abc-af^2-bg^2-ch^2+2fgh)=+,$$

which, putting for shortness,

$$\begin{aligned} &A=x^2+y^2+z^2, \quad B=x'^2+y'^2+z'^2, \quad C=x''^2+y''^2+z''^2, \\ &F=x'x''+y'y''+z'z'', \quad G=x''x+y''y+z''z, \quad H=xx'+yy'+zz', \end{aligned}$$

is by what precedes expressible in the form

$$\tfrac{1}{6}\iint \left\{A(BC-F^2)+B(CA-G^2)+C(AB-H^2)-(ABC-AF^2-BG^2-CH^2+2FGH)\right\} dm\,dm'$$

$$=\tfrac{1}{3}\iint (ABC-FGH)\,dm\,dm',$$

or, since $\sqrt{BC}>F$, $\sqrt{CA}>G$, $\sqrt{AB}>H$, we have $ABC>FGH$, or $ABC-FGH=+$, and therefore the value of the integral is also positive.

2, Stone Buildings, W.C., *6th March*, 1860.

292.

A THEOREM IN CONICS.

[From the Quarterly Journal of Pure and Applied Mathematics, vol. IV. (1861), pp. 131–133.]

The following theorem is given in Todhunter's *Conic Sections*, [Ed. 7, p. 304], "If ellipses be inscribed in a triangle, each with one focus in a fixed straight line, the locus of the other focus is a conic section through the angular points of the triangle." A focus is the intersection of tangents to the conic from the circular points at infinity; and instead of the circular points at infinity we may substitute any two points whatever. This being so, let the equations of the sides of the triangle be $x = 0$, $y = 0$, $z = 0$, and let a pair of tangents to the curve from the points (α, β, γ), $(\alpha', \beta', \gamma')$ meet in the point (ξ, η, ζ), and the other pair of tangents from the same two points meet in the point (X, Y, Z). I find that we have the very simple relation

$$X\xi : Y\eta : Z\zeta = \alpha\alpha' : \beta\beta' : \gamma\gamma',$$

and consequently, when the locus of the point (ξ, η, ζ) is given, that of the point (X, Y, Z) is at once determined by substituting in the equation of the first-mentioned locus, in the place of ξ, η, ζ, the values $\dfrac{\alpha\alpha'}{\xi}$, $\dfrac{\beta\beta'}{\eta}$, $\dfrac{\gamma\gamma'}{\zeta}$, or as we may express it, the second locus is derived from the first by the method of reciprocal trilinear substitutions. And, in particular, when the first locus is a line, the second locus is a conic through the angular points of the triangle, which is Mr Todhunter's theorem. I have con-sidered some of the properties of this substitution in a Memoir "Sur quelques transmutations des Courbes," *Liouville*, t. xiv. (1849), pp. 40–46 and t. xv. (1850), pp. 351–356, [80 and 81].

To demonstrate the theorem, I take for the equation of the conic

$$\sqrt{(lx)} + \sqrt{(my)} + \sqrt{(nz)} = 0,$$

and I write for shortness

$$\begin{aligned}\beta\zeta-\gamma\eta,\ \gamma\xi-\alpha\zeta,\ \alpha\eta-\beta\xi&=A,\,B,\,C,\\ \beta'\zeta-\gamma'\eta,\ \gamma'\xi-\alpha'\zeta,\ \alpha'\eta-\beta'\xi&=A',\,B',\,C',\\ \xi(\beta\gamma'-\beta'\gamma)+\eta(\gamma\alpha'-\gamma'\alpha)+\zeta(\alpha\beta'-\alpha'\beta)&=\Delta,\\ \xi\alpha\alpha'(\beta\gamma'-\beta'\gamma)+\eta\beta\beta'(\gamma\alpha'-\gamma'\alpha)+\zeta\gamma\gamma'(\alpha\beta'-\alpha'\beta)&=\square,\end{aligned}$$

so that in fact

$$\Delta=\begin{vmatrix}\xi&\eta&\zeta\\ \alpha&\beta&\gamma\\ \alpha'&\beta'&\gamma'\end{vmatrix},\qquad \square=\alpha\beta\gamma\alpha'\beta'\gamma'\begin{vmatrix}\dfrac{1}{\alpha}&\dfrac{1}{\beta}&\dfrac{1}{\gamma}\\ \dfrac{1}{\alpha'}&\dfrac{1}{\beta'}&\dfrac{1}{\gamma'}\\ \dfrac{\xi}{\alpha}&\dfrac{\eta}{\beta}&\dfrac{\zeta}{\gamma}\end{vmatrix},$$

and we have

$$BC'-B'C=\xi\Delta,\quad CA'-C'A=\eta\Delta,\quad AB'-A'B=\zeta\Delta,$$

$$\beta\gamma'BC'-\beta'\gamma B'C=\gamma\alpha'CA'-\gamma'\alpha C'A=\alpha\beta'AB'-\alpha'\beta A'B=\square.$$

The conditions in order that the conic

$$\sqrt{(lx)}+\sqrt{(my)}+\sqrt{(nz)}=0$$

may touch the line through (α,β,γ) and (ξ,η,ζ) is

$$\frac{l}{A}+\frac{m}{B}+\frac{n}{C}=0,$$

and the condition in order that it may touch the line through (α',β',γ') and (ξ,η,ζ) is

$$\frac{l}{A'}+\frac{m}{B'}+\frac{n}{C'}=0,$$

and we thus have

$$l:m:n=\frac{1}{\dfrac{1}{BC'}-\dfrac{1}{B'C}}:\frac{1}{\dfrac{1}{CA'}-\dfrac{1}{C'A}}:\frac{1}{\dfrac{1}{AB'}-\dfrac{1}{A'B}},$$

or, what is the same thing,

$$l:m:n=AA'\xi:BB'\eta:CC'\zeta,$$

which determine the constants in the equation of the conic.

Consider now the tangents to the conic from the point (α,β,γ); if the equation of the tangent is assumed to be

$$px+qy+rz=0,$$

then we have

$$p\alpha+q\beta+r\gamma=0,$$

$$\frac{l}{p}+\frac{m}{q}+\frac{n}{r}=0,$$

and these equations are of course satisfied by $p : q : r = A : B : C$, since the line through (ξ, η, ζ) is a tangent. They are also satisfied by

$$p : q : r = \frac{l}{A} : \frac{m}{B} : \frac{n}{C},$$

as is obvious by substitution, we have therefore

$$\frac{l}{A}x + \frac{m}{B}y + \frac{n}{C}z = 0,$$

or more simply

$$\frac{A'\xi}{\alpha}x + \frac{B'\eta}{\beta}y + \frac{C'\zeta}{\gamma}z = 0,$$

for the equation of the other tangent through (α, β, γ), and we have in like manner

$$\frac{A\xi}{\alpha'}x + \frac{B\eta}{\beta'}y + \frac{C\zeta}{\gamma'}z = 0,$$

for the equation of the other tangent through $(\alpha', \beta', \gamma')$; the last-mentioned two lines intersect in the point X, Y, Z, that is we have

$$X : Y : Z = \frac{BC'}{\beta\gamma'} - \frac{B'C}{\beta'\gamma} : \frac{CA'}{\gamma\alpha'} - \frac{C'A}{\gamma'\alpha} : \frac{AB'}{\alpha\beta'} - \frac{A'B}{\alpha'\beta},$$

or attending to an above-mentioned equation, we have

$$X\xi : Y\eta : Z\zeta = \alpha\alpha' : \beta\beta' : \gamma\gamma',$$

which is the property in question. In the particular case, where the points (α, β, γ), $(\alpha', \beta', \gamma')$ are the foci, the theorem is an immediate consequence of the well-known proposition that the product of the perpendiculars let fall from the two foci upon any tangent of the conic is a constant.

2, Stone Buildings, W.C., *17th March*, 1860.

293.

ON A CERTAIN SYSTEM OF FUNCTIONAL SYMBOLS.

[From the Quarterly Journal of Pure and Applied Mathematics, vol. IV. (1861), pp. 225–230.]

M. Hermite, in the Memoir "Sur la Résolution de l'équation du cinquième degré," *Comptes Rendus*, t. xlvi., (1858), p. 508, has given for equations of the fifth degree a solution by means of elliptic functions, analogous to the well-known trigonometrical solution of (the irreducible case of) the cubic equation. He has for this purpose to consider the effect of transforming certain functions $\phi\omega$, $\psi\omega$, connected with the elliptic functions, by replacing therein the argument ω by a linear fraction of the form $\dfrac{c+d\omega}{a+b\omega}$, where a, b, c, d are integer numbers satisfying the condition

$$ad - bc = 1,$$

and he finds for $\phi\left(\dfrac{c+d\omega}{a+b\omega}\right)$ (and the like would apply to $\psi\left(\dfrac{c+d\omega}{a+b\omega}\right)$), six different expressions according to the evenness, or values in relation to the modulus 2, of the coefficients a, b, c, d, viz., the congruence

$$ad - bc \equiv 1 \pmod 2$$

admits of the six solutions shown by the table

	a	b	c	d
I.	1	0	0	1
II.	0	1	1	0
III.	1	1	0	1
IV.	1	1	1	0
V.	1	0	1	1
VI.	0	1	1	1

(where 0 or 1 denotes that the coefficient to which it refers is even or odd), and to each solution there corresponds a distinct value of $\phi\left(\dfrac{c+d\omega}{a+b\omega}\right)$ or $\psi\left(\dfrac{c+d\omega}{a+b\omega}\right)$.

It appeared to me that the formulae in question were not only of the highest importance in the theory of elliptic functions, but that they were very remarkable in and for themselves, and that the question might be looked at in the following manner.

Suppose that A, B are functional symbols, or, so to term them, substitutions, the laws of operation being

$$A(X,Y) = \left(\theta\frac{X}{Y},\ \frac{1}{Y}\right),$$

$$B(X,Y) = (Y,\ X),$$

where θ is a constant. The equations mean that the effect of A is to change X into $\theta\dfrac{X}{Y}$, and Y into $\dfrac{1}{Y}$, and the effect of B to change X into Y, and Y into X. We have

$$A^2X = \theta\frac{\theta X/Y}{1/Y} = \theta^2 X, \quad A^2Y = \frac{1}{1/Y} = Y, \quad \text{or} \quad A^2(X,Y) = (\theta^2X,\ Y),$$

$$B^2X = X, \quad B^2Y = Y, \quad \text{or} \quad B^2(X,Y) = (X,Y);$$

thus B is periodic of the second order; and in the particular case $\theta = 1$, A is also periodic of the second order.

This being so, let K denote any one of the compound symbols A^p, BA^p, A^qB, A^qBA^p, &c.; I say that the species of K is determined by connecting with each of the symbols in question, a certain continued fraction as follows:

A^p,	$p+\omega$,
BA^p,	$p-\dfrac{1}{\omega}$,
A^qB,	$-\dfrac{1}{q+\omega}$,
BA^qB,	$-\dfrac{1}{q-\dfrac{1}{\omega}}$,
A^qBA^p,	$p-\dfrac{1}{q+\omega}$,
BA^qBA^p,	$p-\dfrac{1}{q-\dfrac{1}{\omega}}$,
$A^rBA^qBA^p$,	$p-\dfrac{1}{q-\dfrac{1}{r+\omega}}$,
&c.,	$p-\dfrac{1}{q-\dfrac{1}{r-\dfrac{1}{\omega}}}$,

and when the continued fraction is reduced to the form $\dfrac{c+d\omega}{a+b\omega}$, where, of course, $ad-bc=1$, then, that the corresponding symbol K is of the species I. if $a,b,c,d \equiv 1,0,0,1$ (mod2), and in like manner, that the species is II., III., IV., V., or VI., according as a,b,c,d are in regard to the modulus 2 of the form specified in the second, third, fourth, fifth, or sixth line of the foregoing table.

We have then a system of equations which I write in the form

$$K(X,Y)=\begin{cases} \left(\dfrac{X}{Y},\ Y\right) & \text{for } K \text{ of the species I.}\\ \left(\dfrac{Y}{X},\ X\right) & \text{II.}\\ \left(\dfrac{1}{Y},\ \dfrac{X}{Y}\right) & \text{III.}\\ \left(\dfrac{1}{X},\ \dfrac{Y}{X}\right) & \text{IV.}\\ \left(\dfrac{X}{Y},\ \dfrac{1}{Y}\right) & \text{V.}\\ \left(\dfrac{Y}{X},\ \dfrac{1}{X}\right) & \text{VI.}\end{cases}$$

viz., the first line denotes that $K(X,Y)=(\theta^x X,\ \theta^y Y)$, and so for the other lines, the values of the indices x, y being different in the several cases, but I have written the system in the above form, to put in evidence more distinctly the theorem as applied to the most simple case, where $\theta=1$.

It will be sufficient to indicate how the theorem is proved in this particular case: I will suppose that for

$$K=BA^qBA^p\ldots,$$

we have

$$K(X,Y)=\begin{cases} \left(\dfrac{X}{Y},\ Y\right) & \text{for } K \text{ of the species I.}\\ \left(\dfrac{Y}{X},\ X\right) & \text{II.}\\ \left(\dfrac{1}{Y},\ \dfrac{X}{Y}\right) & \text{III.}\\ \left(\dfrac{1}{X},\ \dfrac{Y}{X}\right) & \text{IV.}\\ \left(\dfrac{X}{Y},\ \dfrac{1}{Y}\right) & \text{V.}\\ \left(\dfrac{Y}{X},\ \dfrac{1}{X}\right) & \text{VI.}\end{cases}$$

and let it be inquired what is the effect of the symbol $K'=A^rBA^qBA^p\ldots$, which, since $A^r(X,Y)=(\theta^r X, Y)$, is equivalent to AK or K, according as r is odd or even.

Let $\Theta = \dfrac{c + d\omega}{a + b\omega}$ be the continued fraction which corresponds to K, and $\Theta' = \dfrac{c' + d'\omega}{a' + b'\omega}$ the continued fraction which corresponds to K', then Θ' is deduced from Θ by putting therein $r + \omega$ for ω, or we have

$$\frac{c + d(\omega + r)}{a + b(\omega + r)} = \frac{c' + d'\omega}{a' + b'\omega},$$

or

$$c' = c + dr,\ d' = d, \qquad a' = a + br,\ b' = b,$$

and therefore when r is even, or $\equiv 0$ (mod2), then

$$c' = c,\ d' = d, \quad a' = a,\ b' = b, \quad (\text{mod}2),$$

or K' is of the same species as K. But in this case since $A^r(X, Y) = (X, Y)$ and r is even, we ought to have K' equivalent to K; and hence r being even, the theorem, assumed to be true for K, is also true for K'.

But if r be odd, or $\equiv 1$ (mod2), then

$$c' = c + 1,\ d' = d, \quad a' = a + 1,\ b' = b, \quad (\text{mod}2),$$

whence, according as

K is of species	I., II., III., IV., V., VI., so
K'	V., IV., VI., II., I., III.

But if from $K(X, Y)$ we derive $AK(X, Y)$, we find that of whatever species K may be, the species of AK is identical with the corresponding species of K', and in the case in question, where r is odd, it is clear that we ought to have K' equivalent to AK. Hence in this case also, and therefore generally, if the theorem be true for K it will be also true for K'.

In a very similar manner it may be shown that if the theorem is true for $K' = A^rBA^qBA^p\ldots$, it will be true for $BA^rBA^qBA^p\ldots$, and the foregoing cases include the case where the symbol is of the form BA^rBA^qB or A^rBA^qB (terminating in a B), for it is only necessary to assume that the last index of A is zero; hence the theorem, if true in any case, as it obviously is for $A^0(X, Y) = (X, Y)$, is true universally.

When θ is not equal 1, the only difference is that the terms of $K(X, Y)$ are respectively multiplied by certain powers θ^x, θ^y of θ, and it is not difficult to see that for $K = A^rBA^qBA^p\ldots$ the indices x, y are of the forms

$$x = \mathfrak{c}r + \mathfrak{b}q + \mathfrak{a}p + \ldots,$$

$$y = \mathfrak{c}'r + \mathfrak{b}'q + \mathfrak{a}'p + \ldots,$$

where $\mathfrak{c}$, $\mathfrak{b}$, $\mathfrak{a}$, ..., $\mathfrak{c}'$, $\mathfrak{b}'$, $\mathfrak{a}'$,... have only the values 0, +1, −1, the values of

$\mathfrak{c}$, $\mathfrak{c}'$ depending on the species of $BA^qBA^p\ldots$,

$\mathfrak{b}$, $\mathfrak{b}'$ $\qquad BA^p\ldots$,

&c.

in a manner which can easily be determined. In the particular case, where $\theta^{16}=1$ (as it is in M. Hermite's problem), it is only necessary to know the values of x, y in regard to the modulus 16, and these values can be expressed in terms of the numbers (a,b,c,d) which correspond to the symbol K (this is in itself a remark-able theorem), and I find that (to modulus 16) for K of the species

$$x=\begin{cases} d(c+d)-1 & \text{I.}\\ -ab & \text{II.}\\ c(c-d)-1 & \text{III.}\\ d(d-c)-1 & \text{V.}\\ c(d-c)+1 & \text{IV.}\\ cd & \text{VI.}\end{cases}, \qquad y=\begin{cases} a(a-b)-1 & \text{I.}\\ b(a+b)-1 & \text{II.}\\ ab & \text{III.}\\ -cd & \text{V.}\\ a(b-a)+1 & \text{IV.}\\ -b(b-a)-1 & \text{VI.}\end{cases}$$

This may be shown by induction in like manner as the theorem in regard to the parts independent of θ. Thus K, K' as before, let K be of the species I., and assume that $K(X,Y)=(\theta^xX,\theta^yY)$ where

$$x=d(c+d)-1,\quad y=a(a-b)-1\ (\text{mod}16),$$

then if r be even, since $A^r(X,Y)=(\theta^rX,Y)$, we ought to have $K'(X,Y)=(\theta^{x+r}X,\theta^yY)$. But in this case K' is of the same species with K, viz., of the species I., and by the theorem, $K'(X,Y)=(\theta^{x'}X,\theta^{y'}Y)$ where

$$x'=d'(c'+d')-1,\quad y'=a'(a'-b')-1\ (\text{mod}16);$$

and we ought to have

$$x'=x+r,\quad y'=y\ (\text{mod}16),$$

that is

$$d'(c'+d')-1=d(c+d)-1+r,\quad (\text{mod}16),$$

$$a'(a'-b')-1=a(a-b)-1,$$

or substituting for a', b', c', d' their values $a+br$, b, $c+dr$, d, these become

$$d(c+d+dr)=d(c+d)+r,\quad (\text{mod}16),$$

$$(a+br)(a-b+br)=a(a-b),$$

or

$$(d^2-1)r=0,$$

$$(\text{mod}16).$$

But r being even, these become

$$d^2 - 1 \equiv 0, \quad 2ab - b^2 + b^2 r \equiv 0 \text{ (mod8)};$$

or, since in the case under consideration, $a, b, c, d \equiv 1, 0, 0, 1$ (mod2), and therefore $b^2 r \equiv 0$ (mod8), the two congruences become

$$d^2 - 1 \equiv 0, \quad a^2 - (a-b)^2 \equiv 0 \text{ (mod8)},$$

which are satisfied since d, a, and $a - b$ are all odd. And in a similar way the theorem is proved for all the other cases.

In M. Hermite's problem, X, Y are replaced by $\phi\omega$, $\psi\omega$, AX, AY are $\phi(\omega+1)$, $\psi(\omega+1)$, and BX, BY are $\phi\left(-\dfrac{1}{\omega}\right)$, $\psi\left(-\dfrac{1}{\omega}\right)$, the properties of the functions ϕ, ψ being such that, as in the preceding investigation,

$$A(X, Y) = \left(\theta\frac{X}{Y}, \frac{1}{Y}\right), \quad B(X, Y) = (Y, X),$$

where $\theta\,(= e^{\frac{1}{8}\pi i})$ is a sixteenth root of unity, and KX, KY are respectively equal to $\phi\left(\dfrac{c + d\omega}{a + b\omega}\right)$, $\psi\left(\dfrac{c + d\omega}{a + b\omega}\right)$, and the formulae are precisely similar to those of the present paper.

2, Stone Buildings, W.C., *5 May*, 1860.

294.

ON A NEW ANALYTICAL REPRESENTATION OF CURVES IN SPACE.

[From the Quarterly Journal of Pure and Applied Mathematics, vol. V. *(1862), pp. 81–86.]*

The employment of a new kind of coordinates for the analytical representation of curves in space is suggested in my former paper under the same title *Journal*, t. III., pp. 225–236 (1859). The idea was as follows: viz. if (x, y, z, w) are current coordinates of a point in space (ordinary point coordinates), and $(\alpha, \beta, \gamma, \delta)$ the coordinates of a particular point, then taking (p, q, r, s, t, u) to represent the minor determinants formed out of the matrix

$$\begin{pmatrix} x & y & z & w \\ \alpha & \beta & \gamma & \delta \end{pmatrix},$$

viz.

$$\begin{aligned} p &= \gamma y - \beta z, \quad & s &= \delta x - \alpha w, \\ q &= \alpha z - \gamma x, \quad & t &= \delta y - \beta w, \\ r &= \beta x - \alpha y, \quad & u &= \delta z - \gamma w, \end{aligned}$$

values which satisfy identically

$$ps + qt + ru = 0,$$

then the equation of a cone passing through a given curve and having for its vertex the arbitrary point $(\alpha, \beta, \gamma, \delta)$, is of the form

$$V = 0,$$

V being a homogeneous function of the six new coordinates (p, q, r, s, t, u). And it was proposed to consider $V = 0$ as the equation of the curve.

But as remarked in the paper, it is not every function V of the coordinates (p, q, r, s, t, u) which equated to zero, does in fact represent a curve. In order that the equation $V = 0$ may represent a curve, it is necessary, that when any infinitesimal variations whatever are given to the constants $(\alpha, \beta, \gamma, \delta)$, thus converting the equation into $V + \delta V = 0$, the two equations $V = 0$, $\delta V = 0$ (considered as equations in ordinary point coordinates) shall represent one and the same curve, whatever the system of infinitesimal variations attributed to $\alpha, \beta, \gamma, \delta$ may be. Let P, Q, R, S, T, U denote the differential coefficients of V in regard to p, q, r, s, t, u respectively, then the equation $\delta V = 0$, breaks up into the equations

$$\begin{aligned} - Ry + Qz - Sw &= 0, \\ Rx \qquad - Pz - Tw &= 0, \\ -Qx + Py \qquad - Uw &= 0, \\ Sx + Ty + Uz \qquad &= 0, \end{aligned}$$

and the system composed of these four equations and the equation $V = 0$ (considered as equations in ordinary point coordinates) must belong to one and the same curve.

The four equations gave

$$PS + QT + RU = 0,$$

a relation between the differential coefficients of V which must be satisfied either identically or in virtue of the equation $V = 0$. And this relation existing, any two of the four equations lead to the other two. Attending exclusively to the coordinates (p, q, r, s, t, u) and considering (x, y, z, w) as mere arbitrary multipliers, the above equation

$$PS + QT + RU = 0$$

is the only relation between the differential coefficients of V which is deducible from the four equations.

But it was noticed that the equation $V = 0$, even when V is a function such that we have (identically or in virtue of the equation $V = 0$) the equation $PS + QT + RU = 0$, does not of necessity represent a curve. Some further relation or relations between the differential coefficients of V must therefore exist, either identically or in virtue of the equation $V = 0$; and such relations can be found by resorting to the second differential $\delta^2 V$ of the function V. In fact not only the equation $\delta V = 0$ but the entire series of relations $\delta^2 V = 0$, $\delta^3 V = 0, \ldots$ should be satisfied by the coordinates of any point of the curve. I find by means of the equation $\delta^2 V = 0$ a plexus of equations, which are consequently necessary, and I am inclined to believe sufficient, in order that the equation $V = 0$ may in fact represent a curve; the equations of the plexus are, it will be seen, very numerous, and certainly only a small number of them are independent, but this is a question which I have not as yet investigated.

Attending to the expressions for p, q, r, s, t, u, we have

$$\begin{aligned} d\alpha &= \qquad - yd_r + zd_q - wd_s = (1), \\ d\beta &= xd_r \qquad - zd_p - wd_t = (2), \\ d\gamma &= -xd_q + yd_p \qquad - wd_u = (3), \\ d\delta &= xd_s + yd_t + zd_u \qquad = (4), \end{aligned}$$

and writing for convenience $\mathsf{a}, \mathsf{b}, \mathsf{c}, \mathsf{d}$ instead of $d_\alpha, d_\beta, d_\gamma, d_\delta$, we have

$$d = (1)\mathsf{a} + (2)\mathsf{b} + (3)\mathsf{c} + (4)\mathsf{d}.$$

It was in effect by operating on V with this symbol and equating to zero the coefficients of $\mathsf{a}, \mathsf{b}, \mathsf{c}, \mathsf{d}$, that the before-mentioned equations

$$\begin{aligned} - Ry + Qz - Sw &= 0, \\ Rx \quad\quad - Pz - Tw &= 0, \\ -Qx + Py \quad\quad - Uw &= 0, \\ Sx + Ty + Uz \quad\quad &= 0, \end{aligned}$$

were found.

If to these equations we join the equation

$$Ax + By + Cz + Dw = 0,$$

where A, B, C, D are arbitrary multipliers, we can express x, y, z, w in terms of A, B, C, D in such manner as to satisfy the four equations, viz. we have

$$\begin{aligned} x &= \quad\quad BU - CT + DP, \\ y &= -AU \quad\quad + CS + DQ, \\ z &= AT - BS \quad\quad + DR, \\ w &= -AP - BQ - CR. \end{aligned}$$

and if in the expressions for $(1), (2), (3), (4)$ we substitute for x, y, z, w these values, and form therewith the value of d, which value I will for distinction call $\mathfrak{D}$, we have

$$\begin{aligned} \mathfrak{D} = \big(&Ud_r + Td_q + Pd_s,\ Qd_s - Sd_q,\ Rd_s - Sd_r,\ Rd_q - Qd_r\big) \\ \big(&Pd_t - Td_p,\ Ud_r + Qd_t + Sd_p,\ Rd_t - Td_r,\ Pd_r - Rd_p\big) \\ \big(&Pd_u - Ud_p,\ Qd_u - Ud_q,\ Rd_u + Sd_p + Td_q,\ Qd_p - Pd_q\big) \\ \big(&Td_u - Ud_t,\ Ud_s - Sd_u,\ Sd_t - Td_s,\ Pd_s + Qd_t + Rd_u\big)\ (A, B, C, D)(\mathsf{a}, \mathsf{b}, \mathsf{c}, \mathsf{d}), \end{aligned}$$

viz. $\mathfrak{D}$ is a lineo-linear function of the two sets of indeterminate quantities (A, B, C, D), $(\mathsf{a}, \mathsf{b}, \mathsf{c}, \mathsf{d})$, the coefficients thereof being the operators

$$Ud_r + Td_q + Pd_s,\ Qd_s - Sd_q,\ \&c.$$

It may be remarked that we have identically

$$\mathfrak{D}V = (PS + QT + RU)(A\mathsf{a} + B\mathsf{b} + C\mathsf{c} + D\mathsf{d}),$$

since obviously each term such as $(Qd_s - Sd_q)V$, which is equal to $QS - SQ$, vanishes identically. The equation $\mathfrak{D}V = 0$ gives therefore only the before-mentioned equation $PS + QT + RU = 0$, which is as it should be.

The equation $\mathfrak{D}^2 V = 0$, is then to be satisfied independently of the values of (A, B, C, D) and $(\mathsf{a}, \mathsf{b}, \mathsf{c}, \mathsf{d})$, and as $\mathfrak{D}$ contains 16 distinct terms, $\mathfrak{D}^2$ will contain in all $\frac{1}{2} \cdot 16 \cdot 17$ or 136 distinct terms. The equation $\mathfrak{D}^2 V = 0$ gives therefore a plexus of 136 equations, and the equations in each succeeding plexus, involved in $\mathfrak{D}^3 V = 0$, $\mathfrak{D}^4 V = 0$, &c. will, of course, be still be more numerous.

If $V = 0$ be the plane conic which is the intersection of the surfaces

$$x^2 + y^2 + z^2 + w^2 = 0,$$

$$ax + by + cz + dw = 0,$$

then we have

$$V = \left(b^2 + c^2,\ -ab,\ -ac,\ \cdot,\ cd,\ -bd,\ c^2 + a^2,\ -bc,\ -cd,\ \cdot,\ ad,\ a^2 + b^2,\ bd,\ -ad,\ \cdot,\ b^2 + d^2,\ ab,\ ac,\ a^2 + d^2,\ bc,\ c^2 + d^2\right)(p, q, r, s, t, u)^2.$$

The values of P, Q, R, S, T, U (omitting a common factor 2) are

$$P = (b^2 + c^2,\ -ab,\ -ac,\ \cdot,\ +cd,\ -bd \mid p, q, r, s, t, u),$$

&c.,

and if we proceed to form a term in $\mathfrak{D}^2 V$, say the coefficient of $A^2 \mathsf{a}^2$, this is $(Ud_r + Td_q + Pd_s)^2 V$, or

$$(a^2 + b^2,\ c^2 + a^2,\ a^2 + d^2,\ -cd,\ +bd,\ -cb \mid U, T, P)^2.$$

The coefficient therein of θ is

$$(a^2 + b^2,\ c^2 + a^2,\ a^2 + d^2,\ -cd,\ +bd,\ -cb \mid -bd,\ cd,\ b^2 + d^2)^2,$$

that is, it is

$$\begin{aligned} &(a^2 + b^2)\, b^2 d^2 - 2cd \cdot cd(b^2 + d^2) \\ &\quad + (c^2 + a^2)\, c^2 d^2 + 2bd \cdot -bd(b^2 + c^2) \\ &\quad + (a^2 + d^2)\,(b^2 + c^2)^2 - 2cb \cdot -bd \cdot cd, \end{aligned}$$

where the terms in which $(b^2 + c^2)$ does not appear as a factor are together equal to

$$a^2 d^2 (b^2 + c^2) + d^2 (b^4 + c^4),$$

the entire expression thus divides by $b^2 + c^2$ the quotient being

$$(a^2 + d^2)(b^2 + c^2) - 4c^2 d^2 - 2b^2 d^2 + a^2 d^2 + d^2 (b^2 + c^2),$$

which is equal to $a^2(b^2 + c^2 + d^2)$, or restoring the factor $b^2 + c^2$, we see that in $\mathfrak{D}^2 V$ the coefficient of $A^2 \mathsf{a}^2$ is

$$a^2 (b^2 + c^2 + d^2)(b^2 + c^2) V + \&c.$$

The complete value must, it is clear, be of the form

$$a^2(b^2+c^2+d^2)V + k(ps+qt+ru),$$

vanishing in virtue of the equations $V=0$, $ps+qt+ru=0$, and this being so, observing that V contains no term in ps, we have $k =$ coefficient ps in

$$(a^2+b^2,\ c^2+a^2,\ a^2+d^2,\ -cd,\ +bd,\ -cb \mid U, T, P)^2,$$

that is

$$k = 2(a^2+b^2,\ c^2+a^2,\ a^2+d^2,\ -cd,\ +bd,\ -cb \mid -bd,\ cd,\ b^2+d^2)(ca,\ ba,\ 0),$$

or

$$\begin{aligned}\tfrac{1}{2}k = (a^2+b^2)\cdot -bd\cdot ca - cd\,[cd\cdot 0 + (b^2+d^2)\,ba]\\ + (c^2+a^2)\cdot cd\cdot ba + bd\,[(b^2+d^2)\,ca - bd\cdot 0]\\ + (a^2+d^2)\cdot 0 - cb\,[-bd\cdot ba + cd\cdot ca],\end{aligned}$$

which is

$$= abcd\Big[-(a^2+b^2)-(b^2+c^2)+(d^2+a^2)+(b^2+d^2)+(b^2-c^2)\Big] = 0.$$

The coefficient k consequently vanishes, and therefore in $\mathfrak{D}^2V$ the coefficient of $A^2\mathsf{a}^2$ is $a^2(b^2+c^2+d^2)V$, but I have not worked out the coefficients of the other terms.

2, Stone Buildings, W.C., *30th October*, 1860.

295.

ON THE CONSTRUCTION OF THE NINTH POINT OF INTER-SECTION OF THE CUBICS WHICH PASS THROUGH EIGHT GIVEN POINTS.

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I REPRODUCE with additional developments the solution which has been given of this interesting problem.

The equation of a given cubic may be written

$$qU - pV = 0,$$

where $U = 0$, $V = 0$ are any two conics meeting the cubic in the same four points; $p = 0$ is the line joining the remaining two points of intersection of the cubic with the conic $U = 0$; and $q = 0$ is the line joining the remaining two points of inter-section of the cubic with the conic $V = 0$; the relation between the arbitrary constant factors implicitly contained in the functions qU and pV is assumed to be properly determined.

The form is employed by Plücker, *Theorie der algebraischen Curven*, p. 56 (1839), and in connexion therewith he gives some geometrical considerations which, he remarks, contain implicitly the solution of the above-mentioned problem.

The form is also the analytical basis of the investigations of M. Chasles, "Construction de la courbe du troisième ordre par neuf points," *Comptes Rendus*, t. xxxvi. (1853), pp. 942–952. In fact calling to mind the theorem that in the pencil of conics $U - aV = 0$ (where a is an arbitrary multiplier) the anharmonic ratio of the multipliers a is equal to the anharmonic ratio of the tangents at any one of the points of intersection, or (what is the same thing) to the anharmonic ratio of the polars of any point whatever in regard to the conics, and recollecting that the anharmonic ratio in question is said to be the anharmonic ratio of the conics them-selves; then since the equation $qU - pV = 0$ is satisfied by the system

$$U - aV = 0, \quad p - aq = 0,$$

(where a is arbitrary) we have at once the theorem, p. 949, viz. that if there be a pencil of lines through a point, and corresponding anharmonically thereto, a pencil of conics through the same four points; the locus of the intersections of a line by the corresponding conic is a cubic through the five points: and, conversely, that a given cubic may be so generated, the point of the pencil of lines, and the four points of the pencil of conics being any five points whatever of the cubic.

This gives at once the construction (M. Chasles' first construction) for the cubic through nine given points. In fact if the points are called 1, 2, 3, 4, 5, 6, 7, 8, 9; then grouping the points in any manner, it is only necessary to find a point x such that

$$x(1, 2, 3, 4, 5) = 6789(1, 2, 3, 4, 5),$$

that is, such that the pencil of lines $x1$, $x2$, $x3$, $x4$, $x5$ shall correspond anharmonically to the pencil of conics 67891, 67892, 67893, 67894, 67895. The foregoing notation is that employed in M. de Jonquières' "Essai sur la génération des Courbes géométriques, &c.," Mém. Sav. Strang., t. xvi. (1858), which I take the opportunity of referring to. In fact, if x satisfies the foregoing condition, then taking through the point x any other line, and corresponding anharmonically thereto a conic through the points 6, 7, 8, 9, the locus of the intersections of the line and conic will be a cubic through the nine points. But the condition in question gives

$$x(1, 2, 3, 4) = 6789(1, 2, 3, 4),$$

$$x(1, 2, 3, 5) = 6789(1, 2, 3, 5),$$

which (by the anharmonic property of the points of a conic) show, the first that x is in a certain conic passing through 1, 2, 3, 4, and the second that x is in a certain conic passing through 1, 2, 3, 5; the two conics intersect in the points 1, 2, 3, and in a fourth point which is the required point x. Or we may say that x is given by the condition that the pencils $x(1, 2, 3, 4)$ and $x(1, 2, 3, 5)$ shall have given anharmonic ratios. It will presently be seen how x can be determined by the ruler alone.

Suppose now that the points 1, 2, 3, 4, 5, 6, 7, 8, 9 are the points of intersection of two cubics; the construction should become indeterminate; this is only the case when the two conics which by their intersection should determine x become one and the same conic. This implies that the conic $x(1, 2, 3, 4)$ passes through 5, or that $x, 1, 2, 3, 4, 5$ are points of the same conic. And then since by the anharmonic property of the points of a conic $x(1, 2, 3, 4) = 5(1, 2, 3, 4)$, we have

$$5(1, 2, 3, 4) = 6789(1, 2, 3, 4).$$

The grouping of the nine points is altogether arbitrary, hence there are in all $(9 \times 70 =)\,630$ such equations, which are really equivalent to only two equations, and which when eight of the points are given, determine the ninth point. Supposing that the given points are 1, 2, 3, 4, 5, 6, 7, 8, the equations for the determination of the remaining point 9 may be taken to be

$$9(5,6,7,8) = 1234(5,6,7,8),$$

$$9(4,6,7,8) = 1235(4,6,7,8),$$

which (it is to be remarked) determine 9 in a similar way to that in which x is given in the construction of the cubic through nine points; viz. 9 is the fourth intersection of two conics which pass through the points 5, 6, 7, 8 and the points 4, 6, 7, 8 respectively. Or we may say that 9 is given by the conditions that the pencils $9(5,6,7,8)$ and $9(4,6,7,8)$ shall have given anharmonic ratios.

The foregoing equations

$$9(5,6,7,8) = 1234(5,6,7,8),$$

$$9(4,6,7,8) = 1235(4,6,7,8),$$

are equivalent to and constitute the geometrical interpretation of the equations obtained (previous to M. Chasles' Memoir) by Weddle in the paper "On the construction of the ninth point of intersection of two curves of the third degree when the other eight points are given," *Cambridge and Dublin Mathematical Journal*, t. vi., pp. 83–86 (1851). In fact, reproducing his analysis with only a slight change of notation, let 012 denote the determinant

$$\begin{vmatrix} x_0 & y_0 & z_0 \\ x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \end{vmatrix}$$

so that $012 = 0$ is the equation of the line through the points 1 and 2; and in like manner let 012345 denote the determinant

$$\begin{vmatrix} x_0^2 & y_0^2 & z_0^2 & y_0z_0 & z_0x_0 & x_0y_0 \\ x_1^2 & \&c. & & & & \end{vmatrix}$$

so that $012345 = 0$ is the equation of the conic through the points 1, 2, 3, 4, 5. Of course 123, 123456, &c. will denote given functions of the coordinates of the points 1, 2, 3, the points 1, 2, 3, 4, 5, 6, &c. This being so

$$012345 \cdot 078 = \lambda \cdot 012347 \cdot 058$$

is the equation of a particular cubic passing through the points 1, 2, 3, 4, 5, 7, 8, and which if we properly determine λ, viz. if we write

$$\lambda = \frac{612345 \cdot 678}{612347 \cdot 658},$$

will also pass through the point 6.

And similarly

$$012345 \cdot 076 = \mu \cdot 012347 \cdot 056$$

is the equation of a particular cubic curve passing through the points 1, 2, 3, 4, 5, 6, 7 and which if we properly determine μ, viz. if we write

$$\mu = \frac{812345 \cdot 876}{812347 \cdot 856},$$

will also pass through the point 8. Hence the two curves, each of them passing through the points 1, 2, 3, 4, 5, 6, 7, 8 will intersect in the remaining point 9; and writing 9 for 0, and combining the two equations, we have

$$\frac{978 \cdot 956}{958 \cdot 976} = \frac{\lambda}{\mu} = \frac{612345}{612347} \cdot \frac{812347}{812345},$$

or, what is the same thing,

$$\frac{956 \cdot 978}{958 \cdot 967} = \frac{123456 \cdot 123478}{123458 \cdot 123467},$$

which is Weddle's equation, and is equivalent to the above-mentioned equation

$$9(5, 6, 7, 8) = 1234(5, 6, 7, 8).$$

To prove this I remark that we have identically

$$012 \cdot 034 \cdot 514 \cdot 523 - 014 \cdot 023 \cdot 512 \cdot 534 = 012345.$$

In fact the left-hand side equated to zero is the equation of the conic through 1, 2, 3, 4, 5, and such left-hand side must therefore, save to a mere numerical factor, be equal to 012345. And to determine this factor it is to be observed that 012345 contains the term

$$+x_0^2 \cdot y_1^2 \cdot z_2^2 \cdot y_3 z_3 \cdot z_4 x_4 \cdot x_5 y_5,$$

but that there is no such term in $012 \cdot 034 \cdot 514 \cdot 523$, and that there is in $-014 \cdot 023 \cdot 512 \cdot 534$ the equivalent term

$$+x_0^2 \cdot y_1^2 \cdot z_2^2 \cdot y_3 z_3 \cdot z_4 x_4 \cdot x_5 y_5,$$

so that the numerical factor is rightly determined.

The foregoing identity written under the form

$$\frac{012 \cdot 034}{014 \cdot 023} - \frac{512 \cdot 534}{514 \cdot 523} = \frac{012345}{014 \cdot 023 \cdot 514 \cdot 523}$$

shows that, when $012345 = 0$, i.e. if 0 be a point of the conic through 1, 2, 3, 4, 5, then we have

$$0(1, 2, 3, 4) = 5(1, 2, 3, 4),$$

which is in fact the anharmonic property of the points of a conic. And observing that $012345 = 051234$, and substituting 5, 6, for 0, 5 respectively, the identity becomes

$$\frac{512 \cdot 534}{514 \cdot 523} - \frac{612 \cdot 634}{614 \cdot 623} = \frac{561234}{514 \cdot 523 \cdot 614 \cdot 623}.$$

The equation $012345 = 0$ may be written

$$\frac{012 \cdot 034}{014 \cdot 023} - \frac{512 \cdot 534}{514 \cdot 523} = 0,$$

and hence the anharmonic ratio of the conics

$$012345 = 0,\ 012346 = 0,\ 012347 = 0,\ 012348 = 0$$

is equal to that of the quantities

$$\frac{512 \cdot 534}{514 \cdot 523},\quad \frac{612 \cdot 634}{614 \cdot 623},\quad \frac{712 \cdot 734}{714 \cdot 723},\quad \frac{812 \cdot 834}{814 \cdot 823},$$

or calling these quantities for a moment $\alpha, \beta, \gamma, \delta$, it is

$$\frac{(\alpha - \beta)(\gamma - \delta)}{(\alpha - \delta)(\beta - \gamma)},$$

where

$$\alpha - \beta = \frac{512 \cdot 534}{514 \cdot 523} - \frac{612 \cdot 634}{614 \cdot 623} = \frac{561234}{514 \cdot 523 \cdot 614 \cdot 623},$$

and forming in this manner the expressions of each of the four factors $\alpha - \beta$, $\gamma - \delta$, $\alpha - \delta$, $\beta - \gamma$, we have

$$\frac{(\alpha - \beta)(\gamma - \delta)}{(\alpha - \delta)(\beta - \gamma)} = \frac{561234 \cdot 781234}{581234 \cdot 671234},$$

so that in the equation

$$\frac{956 \cdot 978}{958 \cdot 967} = \frac{561234 \cdot 781234}{581234 \cdot 671234},$$

the right-hand side is

$$= 1234(5, 6, 7, 8),$$

and since by what precedes the left-hand side is $= 9(5, 6, 7, 8)$, the equation is

$$9(5, 6, 7, 8) = 1234(5, 6, 7, 8),$$

which is the transformation in question.

Now resuming the two equations

$$9(5, 6, 7, 8) = 1234(5, 6, 7, 8),$$

$$9(4, 6, 7, 8) = 1235(4, 6, 7, 8),$$

the right-hand sides are given anharmonic ratios, and as we have seen the question is to find 9 so that the anharmonic ratios $9(5, 6, 7, 8)$, $9(4, 6, 7, 8)$ shall have given values. But for the geometrical solution by the ruler alone, we have the preliminary question, from the given eight points, without the assistance of the before-mentioned conics, to construct the given anharmonic ratios $1234(5, 6, 7, 8)$ and $1235(4, 6, 7, 8)$. The solution of both questions is given in Dr Hart's paper, "Construction by the ruler alone to determine the ninth point of intersection of two curves of the third degree," *Cambridge and Dublin Mathematical Journal*, t. vi., pp. 181, 182 (1851).

The Preliminary Question. The anharmonic ratio $1234(5,6,7,8)$ is equal to that of the polars of an arbitrary point X in regard to the conics 12345, 12346, 12347, 12348 respectively (these polars all pass through one and the same point). Now to construct the polars of X in regard to these conics, and first in regard to the conic 12345: the fourth harmonics of X in regard to the lines 12, 34, in regard to the lines 13, 42, and in regard to the lines 14, 23, meet in a point; and considering the several combinations 1234, 1235, 1245, 1345, 2345 we have thus five points; these lie on a line which is the required polar of X in regard to the conic 12345. The polars in regard to the other conics are obtained in the same manner; and it is clear that the first above-mentioned point (viz. that deduced from the points 1, 2, 3, 4) is in fact the point of intersection of the four polars, or point of the pencil formed by the polars.

The Principal Question then is, given the points 4, 5, 6, 7, 8 to find the point 9, such that

$$9(5,6,7,8) = 1234(5,6,7,8),$$

$$9(4,6,7,8) = 1235(4,6,7,8),$$

where the right-hand sides represent given anharmonic ratios.

For this, let 65, 74 (see the figure) meet in M, and on 74 find a point Q such

[Figure: pencils through points 4, 5, 6, 7, 8 with auxiliary points M, N, Q, R, K, L used to construct the ninth intersection point 9.]

that the anharmonic ratio of the points $(4,M,7,Q)$ may be equal to the given ratio $1235(4,6,7,8)$, say

$$(4,M,7,Q) = 1235(4,6,7,8),$$

and say let 64, 85 meet in N and on 85 find a point R, such that the anharmonic ratio of the points $5,N,R,8$ is equal to the given anharmonic ratio $1234(5,6,7,8)$,

$$(5,N,R,8) = 1234(5,6,7,8).$$

Join QR meeting 65 in K and 64 in L; then $7K$ and $8L$ will meet in the required point 9.

In fact taking Y as the intersection of the lines $65KM$ and $8L$, and Z as the intersection of the lines $64NL$ and $7K$; then, as is clear from the figure, first, the anharmonic ratio $9(4,6,7,8)$ is equal to that of the points $(4,6,Z,L)$ on the line $46ZL$, that is

$$9(4,6,7,8) = (4,6,Z,L),$$

which is

$$= (4,M,7,Q),$$

since the lines $6M$, ZL, LQ meet in the point K; but by the construction $(4,M,7,Q) = 1235(4,6,7,8)$, that is, we have

$$9(4,6,7,8) = 1235(4,6,7,8);$$

and, secondly, the anharmonic ratio $9(5,6,7,8)$ is equal to that of the points $(5,6,K,Y)$ on the line $56KY$, that is

$$9(5,6,7,8) = (5,6,K,Y),$$

which is

$$= (5,N,R,8),$$

since the lines $6N$, KR, $Y8$ meet in the same point L; but by the construction $(5,N,R,8) = 1234(5,6,7,8)$, that is, we have

$$9(5,6,7,8) = 1234(5,6,7,8),$$

so that the point 9 satisfies the required conditions.

It has been already remarked that the point x in M. Chasles' theorem for the construction of the cubic through nine points is determined by precisely similar conditions to those which determine the ninth intersection of the two cubics; that is, the foregoing construction by the ruler alone is applicable to the determination of the point x; and when this is once obtained, the remainder of the construction, giving the points of the cubic through the nine given points, can obviously be performed by the ruler alone. The construction for the cubic through nine points gives implicitly the relation between ten points of the cubic and such relation is accordingly expressed by the equation

$$x(1,2,3,4,5,10) = 6789(1,2,3,4,5,10),$$

which is one out of 210 similar forms. But it is possible that some more convenient form of the relation between the ten points may yet be found.

I proceed to further develope the analytical theory. Writing for convenience ω in the place of 10, we have

$$x(1,2,3,4,5,6) = 789\omega(1,2,3,4,5,6),$$

or, what is the same thing,

$$x(1,2,3,4) = 789\omega(1,2,3,4),$$

$$x(1,2,3,5) = 789\omega(1,2,3,5),$$

$$x(1,2,3,6) = 789\omega(1,2,3,6),$$

which belong to three conics each of them passing through 1, 2, 3, and which must have a remaining fourth point of intersection.

The equation of the first conic is

$$012 \cdot 034 = \frac{\lambda}{\mu} \cdot 014 \cdot 023,$$

where

$$\lambda = 789\omega 12 \cdot 789\omega 34,$$

$$\mu = 789\omega 14 \cdot 789\omega 23,$$

and thence, in virtue of an identical equation already referred to,

$$-\lambda - \mu = 789\omega 13 \cdot 789\omega 24.$$

But we have identically

$$123 \cdot 0pq = 1pq \cdot 023 + 2pq \cdot 031 + 3pq \cdot 012,$$

and thence in particular

$$123 \cdot 034 = 134 \cdot 023 + 234 \cdot 031,$$

$$123 \cdot 014 = 214 \cdot 031 + 314 \cdot 012,$$

and the equation of the conic may therefore be written

$$012(134 \cdot 023 + 234 \cdot 031)\,\mu - 023(214 \cdot 031 + 314 \cdot 012)\,\lambda = 0,$$

that is

$$031 \cdot 012 \cdot 234 \cdot \mu + 012 \cdot 023 \cdot 314\,(-\lambda - \mu) + 023 \cdot 031 \cdot 124 \cdot \lambda = 0,$$

or, substituting for μ, $-\lambda-\mu$, λ, their values, this is

$$031 \cdot 012 \cdot 234 \cdot 789\omega 14 \cdot 789\omega 23$$

$$+\,012 \cdot 023 \cdot 314 \cdot 789\omega 13 \cdot 789\omega 42$$

$$+\,023 \cdot 031 \cdot 124 \cdot 789\omega 12 \cdot 789\omega 34 = 0,$$

or, what is the same thing,

$$\frac{234 \cdot 789\omega 14 \cdot 789\omega 23}{023} + \frac{314 \cdot 789\omega 13 \cdot 789\omega 24}{031} + \frac{124 \cdot 789\omega 12 \cdot 789\omega 34}{012} = 0,$$

or, making a slight change of form, the equation of the conic is

$$\frac{423 \cdot 789\omega 41 \cdot 789\omega 23}{023} + \frac{431 \cdot 789\omega 42 \cdot 789\omega 31}{031} + \frac{412 \cdot 789\omega 43 \cdot 789\omega 12}{012} = 0.$$

The equations of the other two conics are deduced by writing successively 5 and 6 in the place of 4; and the condition in order that the conics may have a remaining fourth point of intersection is

$$\begin{vmatrix} 423 \cdot 789\omega 41 & 431 \cdot 789\omega 42 & 412 \cdot 789\omega 43 \\ 523 \cdot 789\omega 51 & 531 \cdot 789\omega 52 & 512 \cdot 789\omega 53 \\ 623 \cdot 789\omega 61 & 631 \cdot 789\omega 62 & 612 \cdot 789\omega 63 \end{vmatrix} = 0.$$

This equation, say $D = 0$, expresses the relation between the coordinates of the ten points 1, 2, 3, 4, 5, 6, 7, 8, 9, ω, of the cubic. Hence if 123456789ω denote the determinant

$$\begin{vmatrix} x_1^3 & y_1^3 & z_1^3 & y_1^2 z_1 & z_1^2 x_1 & x_1^2 y_1 & y_1 z_1^2 & z_1 x_1^2 & x_1 y_1^2 & x_1 y_1 z_1 \\ x_2^3 & \&c. \end{vmatrix},$$

123456789ω must be a factor of D, and a little consideration shows that the other factor which is of the order one as regards the coordinates of each of the points 1, 2, 3, and of the order three as regards the coordinates of each of the points 7, 8, 9, ω, must be of the form $123 \cdot 789 \cdot 78\omega \cdot 79\omega \cdot 89\omega$. We must therefore have

$$D = \epsilon \cdot 123 \cdot 789 \cdot 78\omega \cdot 79\omega \cdot 89\omega \cdot 123456789\omega,$$

where the merely numerical factor ϵ is, I believe, equal to $+1$ or else to -1.

In order to verify the factor $123 \cdot 789 \cdot 78\omega \cdot 79\omega \cdot 89\omega$, observing that the points 7, 8, 9, ω enter symmetrically, it will be sufficient to show that 123, 789 are each of them factors of D, or, what is the same thing, that if $123 = 0$, or if $789 = 0$, then in either case $D = 0$.

First, if $123 = 0$, we may write

$$x_3 = \lambda x_1 + \mu x_2,$$

$$y_3 = \lambda y_1 + \mu y_2,$$

$$z_3 = \lambda z_1 + \mu z_2,$$

equations which give

$$423 = \lambda \cdot 421, \quad 431 = \lambda \cdot 421, \quad \&c.,$$

and the equation $D = 0$, thus becomes

$$\begin{vmatrix} 789\omega 41 & 789\omega 42 & 789\omega 43 \\ 789\omega 51 & 789\omega 52 & 789\omega 53 \\ 789\omega 61 & 789\omega 62 & 789\omega 63 \end{vmatrix} = 0,$$

or, what is the same thing,

$$\begin{vmatrix} 789\omega 14 & 789\omega 24 & 789\omega 34 \\ 789\omega 15 & 789\omega 25 & 789\omega 35 \\ 789\omega 16 & 789\omega 26 & 789\omega 36 \end{vmatrix} = 0.$$

Now if the terms in the same column are multiplied by $789\omega 56$, $789\omega 64$, $789\omega 45$ respectively and added, then for the first column the sum is

$$789\omega 14 \cdot 789\omega 56 + 789\omega 15 \cdot 789\omega 64 + 789\omega 16 \cdot 789\omega 45,$$

which is $= 0$, and the sums for the second and third columns are each $= 0$ in the same manner: wherefore the determinant vanishes as it should do.